Journal of Manufacturing Processes 72 (2021) 544–552

Contents lists available at ScienceDirect

Journal of Manufacturing Processes

journal homepage: www.elsevier.com/locate/manpro

Online performance optimization for complex robotic assembly processes

Heping Chen [a,*], Hongtai Cheng [b]

[a] Ingram School of Engineering, Texas State University, San Marcos, TX 78746, United States of America
[b] Department of Mechanical Engineering and Automation, Northeastern University, China

ARTICLE INFO

Keywords:
Industrial robot
Assembly parameter optimization
Assembly process optimization
Gaussian process regression
Bayesian optimization

ABSTRACT

The variations and uncertainties in complex assembly processes could cause conventional industrial robots failure to perform assemblies. Therefore, intelligent robots must be developed which can adapt to these variations and uncertainties; moreover, the assembly process parameters must be optimized to satisfy the performance requirements such as First Time Throughput (FTT) rate and cycle time. However, it is challenging to optimize the performance of complex robotic assembly systems. The existing solutions are not efficient and human knowledge is needed to perform experiments and explore optimal parameters. These requirements greatly limit the applications of robotic assembly in manufacturing automation. In this paper, a robotic assembly process optimization method is developed for complex assembly processes. We propose a modeling method based on Gaussian Process Regression (GPR) to construct a model to build the relationship between the process parameters (input) and system performance (output). The GPR surrogated Bayesian Optimization Algorithm (GPRBOA) is improved to iteratively optimize the robotic assembly performance. Two industrial assembly processes (a tight-tolerance peg-in-hole assembly process and a torque converter assembly process) are used to verify the proposed method. The experiments were performed many times and the results demonstrate that the proposed GPRBOA method can optimize the complex assembly process parameters effectively and efficiently. The proposed complex assembly process optimization method opens a door for online manufacturing process optimization and will greatly reduce the manufacturing cost.

1. Introduction

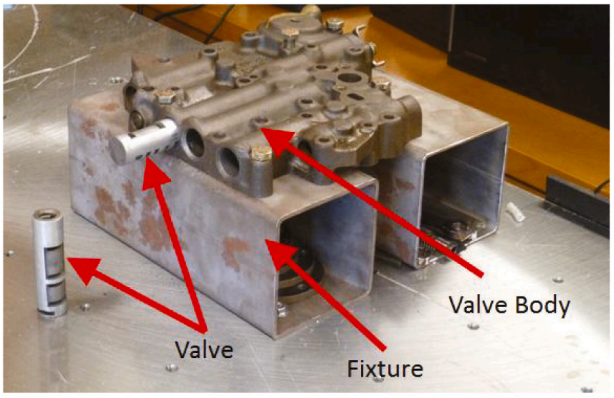
The parts in the manufacturing processes may have variations because they could be produced in different batches and/or by different suppliers. Moreover, the constantly changing environment could cause uncertainties and variations in the manufacturing system. Because of these issues, many assembly processes performed by conventional industrial robots fail quite often [1], [2]. The valve body assembly process and torque converter assembly process as shown in Fig. 1 are examples to have the failure problem, which significantly increases the manufacturing cost and decreases the production efficiency. Thus, a force-controlled intelligent assembly technology is implemented in this paper to perform these assembly processes.

Installing a valve into a valve body is quite difficult even though force control is deployed. The valve radius is about 24.96 mm and that of the valve body is about 25.00 mm. Hence, the assembly tolerance is about 40 μm which is quite close to the robot repeatability. Because of the part location uncertainties caused by fixture errors and part dimensional variations, the valve may be off the hole axis on the valve body. Hence, the valve could be stuck on the valve body because of this misalignment, which will cause this assembly process failure. To mitigate this problem, intelligent method to adjust the assembly parameters must be investigated. The parameters in the valve body assembly process include insertion force, search speed, search force and search radius etc. These parameters greatly influence the robotic assembly performance.

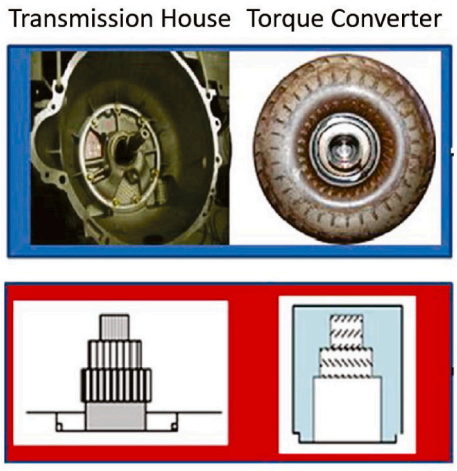
Similarly installing a converter into transmission is also quite challenging. Because the gears in the converter are not fixed, the torque converter must be rotated back and forth in order to engage the gears. Meanwhile, it must be hopped up and down to avoid the gears being stuck together. In the torque converter assembly, there are typically three or four distinct stages to be assembled sequentially. At each stage, there are several parameters to be tuned, such as search amplitude and speed, rotation speed and angle, insertion force and hopping amplitude and frequency. If some of these assembly parameters are not tuned properly, the torque converter assembly process may fail.

* Corresponding author.
E-mail addresses: hc15@txstate.edu (H. Chen), henght@mail.neu.edu.cn (H. Cheng).

https://doi.org/10.1016/j.jmapro.2021.10.047
Received 7 September 2021; Received in revised form 20 October 2021; Accepted 24 October 2021
Available online 5 November 2021
1526-6125/© 2021 The Society of Manufacturing Engineers. Published by Elsevier Ltd. All rights reserved.



(a)



(b)

Fig. 1. a) The valve body assembly; b) torque converter assembly.

Many researchers have studied the optimization of assembly sequence planning and robot assembly cell design etc. [3], [4], [5]. However, the performance of the robotic assembly processes should also be investigated. As mentioned above, there are different assembly process parameters for different assembly processes. These parameters must be tuned carefully in order to successfully perform the assembly processes and achieve minimal cycle time.

Several offline algorithms have been developed to optimize assembly cycle time [6], [7]. Based on random search, Genetic algorithms (GA) was proposed to explore the optimal robotic assembly parameters [7], [8], [9], [10]. Due to the limitations of GA methods such as low efficiency, researchers deployed Artificial Neural Network (ANN) to reduce searching space by categorizing the assembly process parameters [11], [12]. Some researchers proposed the design-of-experiment (DOE) methods [13], [14] to systematically optimize the assembly parameters. After experimental data are collected and analyzed, the most significant parameters are selected and tuned [15], [16], [17], [18], [19]. However, it takes a lot of effort to use these methods to optimize the assembly process parameters. Hence, it is not suitable to use them to perform optimization process in the real production line. To solve the part variation problem, Tomasic et al. [20] developed a mixed-effect model to compensate the variation by measuring geometrical characteristics. Because the extra sensors have to be deployed and some variations may not be corrected using the model, the proposed method may cause assembly failure.

To our best knowledge, there is no practical robotic assembly parameter optimization method proposed to perform assembly performance optimization online. For robotic assembly systems, the robots, assembly processes, system configurations and control strategies etc. could be quite different. Hence, it is almost impossible to build a physical mathematic model to realize the parameter optimization process. To deal with this challenge, a parameter optimization method for robotic assembly processes must be investigated without constructing a physical model.

Using a set of training data, Gaussian Process Regression (GPR), which is a non-parametric method [21], [22], [23], [24], constructs a model between inputs and outputs with system uncertainties and noisy data. Many researchers used the method to model complex systems, such as biomedical application [25], semiconductor manufacturing [26], sensor networks [27], battery applications [28], dynamical systems [29], geographical terrains [30], complex environments [31] and nonlinear systems [32], [33].

The Bayesian Optimization Algorithm (BOA) [34], [35] can be applied to solve the black-box optimization problem. It generates new candidate solutions by utilizing Bayesian Network [36], [37] based on a probability distribution. At each iteration, the computed model is updated using new data samples. Many offline optimization problems [38], [39], [40], [41], [42] has been solved using BOA because it can balance the parameter optimization efficiency and modeling difficulty effectively.

Because GPR can model a complex system and BOA can provide an optimal solution for a complex system, they can be combined to formulate GPR surrogated BOA (GPRBOA) [43], [44], [45], [46], [47] to optimize a complex system. GPRBOA has been applied to solve many problems in machine learning [48], [49], environmental monitoring [50] and material design [38], [45] etc. Therefore, we propose to use GPRBOA to solve the complex robotic assembly problem. However, there are three challenges using GPRBOA in the online robotic assembly process optimization:

- Modeling of complex manufacturing processes using GPR: The prior knowledge used to determine the model is usually not available for the manufacturing processes. There is a lack of reliable mathematical models that directly link manufacturing performance with input parameters, and various types of uncertainties. Hence, how to construct a model online using the data collected step by step?
- Online parameter optimization: Because of the noise, system uncertainties and possible failures of the assembly processes [51], how to optimize the process performance online?
- Local minimum: The optimization process could converge locally. How to generate an optimal solution globally?

The aforementioned challenges towards successfully applying robotic assembly performance optimization methods in manufacturing motivate the proposed research.

In this paper, GPRBOA is implemented to optimize the performance of complex robotic assembly systems online. A model of a complex assembly system is constructed and the assembly process parameters are optimized iteratively. The modeling process and optimization process are balanced online. The valve body assembly process and torque converter assembly process are implemented and the two assemblies were performed to validate the proposed method. The proposed method greatly improves the optimization accuracy and efficiency of complex assembly processes compared to other existing methods. Moreover, because the proposed method is implemented online without interrupting the production line, it greatly decreases the manufacturing cost. The proposed algorithms will advance the state-of-the-art technology to the course of optimal manufacturing process performance.

The remaining paper is organized as follows. Section 2 presents the proposed online assembly parameter optimization algorithm. The experimental results are given in Section 3 and the paper is concluded in Section 4.

2. Proposed solution

In order to control the manufacturing cost and increase robotic assembly efficiency, it is better to perform the optimization process quickly and efficiently. Because the existing methods cannot handle process optimization problem effectively for complex assembly processes, an online parameter learning method is proposed. The general framework of the proposed method is illustrated in Fig. 2.

The robot controller will deploy the potential optimal assembly parameters to perform a complex assembly process. Once the assembly process is done, the performance index will be evaluated, and the collected data will be used as the inputs to the parameter learning method. A new set of parameter candidates will be generated to perform the complex assembly process again. The optimization process will be performed iteratively until the assembly performance requirements are satisfied. Fig. 3 details the assembly process parameter learning method.

After an assembly process is performed using the assembly process parameters (an initial set of parameters at the beginning, otherwise a new set of parameters), the experimental data can be collected, and a model based on GPR can be constructed. If the model is not converged, the exploration method or exploitation method will be implemented to generate a new set of process parameters. To avoid possible local convergence, the exploration method is applied to generate a set of assembly parameters in the unexplored regions. The exploitation method is used to obtain current optimal solution based on the constructed model. The method to balance the exploration and exploitation processes is also investigated. If the constructed model is converged, an optimal set of process parameters is found and the production process will be started.

2.1. Modeling using GPR

GPR can deal with modeling problem for a complex system with uncertainties and observation noise. Thus, GPR is ideal for complex robotic assembly process modeling.

During an assembly process, the input data (assembly process parameters) and output data (performance index) can be collected. Suppose N datasets $(\mathbf{x}_i, y_i)$ (i = 1,…,N) are obtained. The following model can be used to describe the relationship between input data $\mathbf{x}_i$ and output data y_i:

$$y_i = f(\mathbf{x}_i) + \varepsilon_i, \varepsilon_i \sim N(0, \sigma^2) \tag{1}$$

where ε_i is noise with mean 0 and variance σ. Suppose $f(\mathbf{x}_i)$ (i=1,…,N) are random variables with a joint Gaussian distribution, and $f(\mathbf{x}_i)$ has the value of the latent function $f(\mathbf{x})$ at $\mathbf{x}_i$. We can then define a complex assembly process $f(\mathbf{x})$ using its covariance function $k(\mathbf{x}, \mathbf{x}')$ and mean function $m(\mathbf{x})$:

$$f(x) \sim GP(m(x), k(x, x')) \tag{2}$$

where GP represents Gaussian Process; x and $\mathbf{x}'$ are two sets of input variables. Thus, the output covariance function cov(y_i,y_k) of a complex assembly process can be defined as:

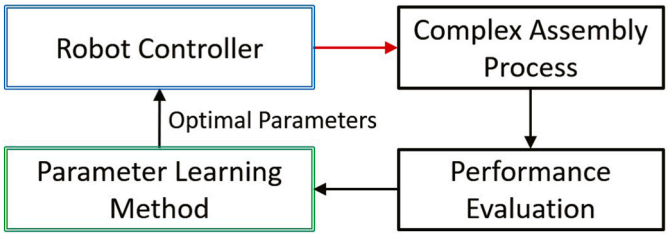


Fig. 2. The general framework of the proposed assembly process optimization method.

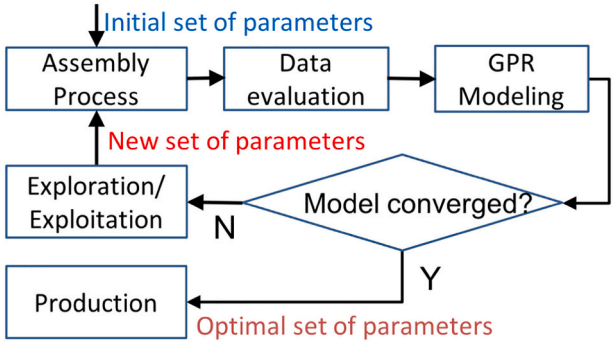


Fig. 3. The optimization process for a complex robotic assembly process based on Gaussian Process Regression and Bayesian Optimization.

$$cov(y_i, y_k) = k(\mathbf{x}_i, \mathbf{x}_k) + \sigma^2 \delta_{ik} \tag{3}$$

δ_{ik}=1 iff $i = k$ and zero otherwise.

Given the observed datasets $(\mathbf{X}, \mathbf{y}) = \{(\mathbf{x}_i, y_i)(i=1,...,N)\}$, a non-parametric model of a complex assembly process can be constructed based on Gaussian Process. The model can be updated by adding more datasets step-by-step. Once a model is available, the performance index (output $\mathbf{y}_p$) can be predicted given a set of assembly process parameters $\mathbf{X}_p$, i.e., $(\mathbf{X}_p, \mathbf{y}_p) = \{(\mathbf{x}_{pj}, y_{pj})(j=1,...,M)\}$. The joint distribution of the two datasets can be calculated using:

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{y}_p \end{bmatrix} \sim N(0, \Lambda) \tag{4}$$

where

$$\Lambda = \begin{bmatrix} K(\mathbf{X}, \mathbf{X}) + \sigma^2 \mathbf{I} & K(\mathbf{X}, \mathbf{X}_p) \\ K(\mathbf{X}_p, \mathbf{X}) & K(\mathbf{X}_p, \mathbf{X}_p) \end{bmatrix} \tag{5}$$

$K(\mathbf{X}', \mathbf{X}^*)$ ($\mathbf{X}'$ and $\mathbf{X}^*$ refer to $\mathbf{X}$ and $\mathbf{X}_p$) is the covariance matrix:

$$K\left(\mathbf{X}', \mathbf{X}^*\right) = \left(k\left(\mathbf{x}_i', \mathbf{x}_j^*\right)\right)_{I,J} \tag{6}$$

where I and J refer to M and N.

The performance index of a complex robotic assembly process can then be predicted by deriving the conditional distribution:

$$\mathbf{y}_p | \mathbf{X}, \mathbf{y}, \mathbf{X}_p \sim N(\Phi(\mathbf{y}_p), \Psi(\mathbf{y}_p)) \tag{7}$$

$$\begin{aligned} \Phi(\mathbf{y}_p) &= K(\mathbf{X}_p, \mathbf{X}) \Omega^{-1} \mathbf{y} \\ \Psi(\mathbf{y}_p) &= K(\mathbf{X}_p, \mathbf{X}_p) - K(\mathbf{X}_p, \mathbf{X}) \Omega^{-1} K(\mathbf{X}, \mathbf{X}_p) \end{aligned} \tag{8}$$

where $\Phi(\mathbf{y}_p)$ is the mean and $\Psi(\mathbf{y}_p)$ the variance of $\mathbf{y}_p$; $\Omega = K(\mathbf{X}, \mathbf{X}) + \sigma^2 I$. Once the mean and variance of the performance index of a complex assembly process are available, its performance can be evaluated. From the above equations, we can see that a non-parametric model of a complex system can be determined if the covariance function $k(\mathbf{x}_i, \mathbf{x}_j)$ is known. Table 1 shows some covariance functions.

The above covariance functions have hyperparameters $\mathbf{H}$. For example, the Squared Exponential function has hyperparameters $\mathbf{H} = [l\ \lambda]$.

Table 1
Examples of covariance functions $k(\mathbf{x}_i, \mathbf{x}_j)$.

Name	Equation
Linear function	$\mathbf{x}_i \cdot \mathbf{x}_j$
Squared exponential function	$\lambda^2 exp\left(-\frlac{\lvert \mathbf{x}_i - \mathbf{x}_j \rvert^2}{2l^2}\right)$
Gaussian radial basis function	$\exp(-\beta \lvert \mathbf{x}_i - \mathbf{x}_j \rvert^2)$
Sigmoid function	$\tanh[\beta(\mathbf{x}_i \cdot \mathbf{x}_j) + c]$
Polynomial function	$[\beta(\mathbf{x}_i \cdot \mathbf{x}_j) + c]^\alpha$

To build a model using a given dataset (**X,y**), we must explore a covariance function $k(\mathbf{x}, \mathbf{x}')$ to fit the data set optimally. For a data set (**X,y**), the posterior probability of $f(\mathbf{x})$ can be computed:

$$p(f|\mathbf{X},\mathbf{y},\mathbf{H}) = \frac{p(\mathbf{y}|\mathbf{X},f,\mathbf{H})p(f|\mathbf{X},\mathbf{H})}{p(\mathbf{y}|\mathbf{X},\mathbf{H})} \tag{9}$$

We can calculate the marginal likelihood using:

$$p(\mathbf{y}|\mathbf{X},\mathbf{H}) = \int p(\mathbf{y}|\mathbf{X},f,\mathbf{H})p(f|\mathbf{X},\mathbf{H})df \tag{10}$$

Because GP constructs a non-parameter model, which means $f(\mathbf{x})$ is not explicitly expressed. $p(\mathbf{y}|\mathbf{X},\mathbf{H})$ refers to the likelihood of **H** in the covariance function. Based on the Gaussian assumption, we can calculate an analytical solution of the marginal likelihood:

$$logp(\mathbf{y}|\mathbf{X},\mathbf{H}) = -\frac{1}{2}\mathbf{y}^{\mathrm{T}}\Omega^{-1}\mathbf{y} - \frac{1}{2}log|\Omega| - \frac{N}{2}log2\pi \tag{11}$$

Hence, the optimal hyperparameters **H** can be obtained by maximizing the marginal likelihood:

$$\mathbf{H}^{*} = \underset{\mathbf{H}}{\mathrm{argmax}}\, logp(\mathbf{y}|\mathbf{X},\mathbf{H}) \tag{12}$$

Different methods can be applied to solve the above optimization problem, such as Nelder-Mead (Simplex) method. An optimal set of hyperparameters can be obtained after optimization. For a new set of assembly process parameters $\mathbf{x}_p$, Eq. (8) can be used to predict the output mean and variance, which can be used to evaluate the assembly process performance.

2.2. Bayesian optimization algorithm (exploitation)

The proposed assembly process optimization algorithm requires to generate new candidate solutions iteratively. In this paper, we implement BOA to evaluate the probability distributions of possible solutions. Once a GP model is generated, the mean and variance of the performance index over the input domain can be predicted which can be used to formulate an acquisition function. By optimizing the acquisition function, optimal assembly performance can be obtained. Hence, the acquisition function determines the new candidate solution; thus, it will affect the quality of the constructed GP model and result of assembly process optimization. Therefore, it is very important to choose a proper acquisition function. Typically, the acquisition function is formulated based on the computed mean and variance in Eq. (8). Different techniques can then be used to determine an optimal candidate such as Upper Confidence Bound (UCB) [52], Probability of Improvement [53] and Expected Improvement [54]. In this paper, we adopt UCB which is defined as:

$$UCB(\mathbf{x})\Delta = \Phi(\mathbf{x}) + \kappa\cdot\Psi(\mathbf{x}) \tag{13}$$

where κ is a scaling factor. By combining the mean $\Phi(\mathbf{x})$ with variance $\Psi(\mathbf{x})$, the search space can be reduced because it can eliminate points that are impossible to be an optimal solution. The assembly performance index can then be optimized by minimizing UCB.

2.3. GPRBOA-RV method (exploration)

GPRBOA updates the model and generates a potential candidate based on the exploration and exploitation processes. The exploitation process uses the proposed UCB method to find a new candidate for the next experiment; while the exploration process produces a new candidate in the unexplored region. The optimization process could generate a local solution if we do not balance the exploitation and exploration processes properly. Therefore a new algorithm called GPRBOA-RV (Random Variation) is proposed to balance these two processes:

$$\mathbf{x}^{*} = \begin{cases} \mathrm{argmin}UCB(\mathbf{x}) & rand(1) > \gamma \\ \mathrm{argmax}RV(\mathbf{x}) & otherwise \end{cases} \tag{14}$$

where γ is an index evaluated using the system performance and hyperparameters; $RV(\mathbf{x})$ is proposed to explore the farthest unexplored region defined as:

$$RV(\mathbf{x}) = \min_{i\in 1\cdots L} d(\mathbf{x}-\mathbf{x}_i) \tag{15}$$

where L is the number of points explored so far; $d(\mathbf{x}-\mathbf{x}_i)$ is the distance between two sets of parameters $\mathbf{x}$ and $\mathbf{x}_i$. If rand(1)>γ, a new set of parameters as shown in Fig. 3 is generated at each iteration by exploiting the current model; otherwise, the unexplored region is investigated to generate a new candidate. Hence, the exploration process avoids the optimization process being converged to a local solution while the exploitation process controls the system towards optimal solution.

3. Experimental results

The proposed method is demonstrated using two force-controlled assembly processes: valve body assembly and torque converter assembly. The hybrid position/force control strategy [55] is used in the assembly processes. The force is controlled along the insertion direction (the *Z*-axis). In order to avoid jamming during the insertion processes, the torques about the X-axis and Y-axis are also controlled. The desired torque is set to be 0.

3.1. Experiments for valve body assembly

Fig. 4 shows the experimental system for a high precision force-controlled assembly system. The assembly system consists of an ABB IRB140 robot with a controller, an ATI force/torque sensor and a vacuum suction tool to hold a valve. The force/torque sensor is installed on the robot end-effector.

The offline and online parameter optimization processes were performed on a computer. The robot controller is connected to the computer via ethernet to transfer data. The assembly process is controlled by the robot controller using the assembly parameters generated by the computer; the computer receives the experimental data such as the assembly cycle time or assembly failure. The computer will update the GPR model of the assembly process and generate a new set of parameters using the exploitation or exploration processes. In order to simulate the variation of the part locations, random noise is added to the part location. Fig. 5 shows a part with location error and the assembled part.

The assembly process is performed using force/torque control including the force/torque-controlled insertion and force guided spiral

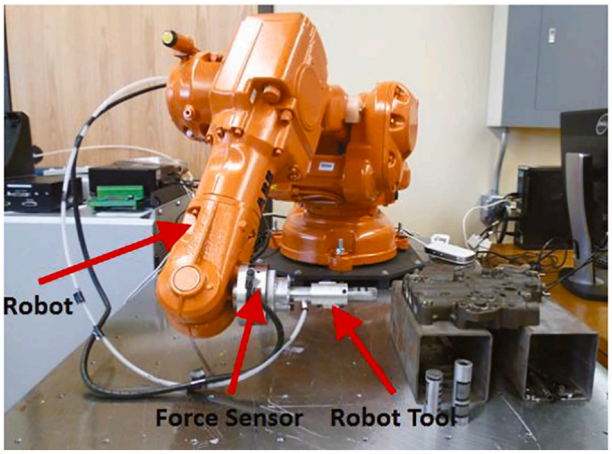


Fig. 4. The experimental system for valve body assembly.

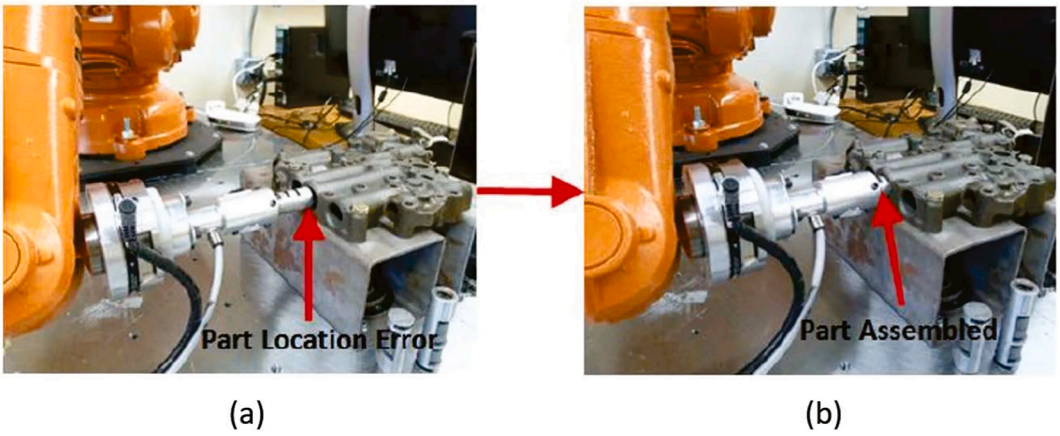


Fig. 5. (a) The part is off the hole; and (b) the part is successfully assembled.

search. There are three assembly parameters to be tuned: Insertion Force (IF), Search Speed (SS) and Search Force (SF). Three groups of experiments listed in Table 2 were performed in order to demonstrate the advantages of the proposed method. The format of the parameters IF, SS and SF is (Minimum Value: Interval: Maximum Value) in Table 2. 3 L represents that three levels of each parameter are used for experiments. IF, for example, has 3 levels (50,75,100) for DOE 3 L. ML means that more than three levels for each parameter are used for experiments. IF, for example, has 11 levels (from 50 to 100 with the step size 5) for GPRBOA-RV ML. For the proposed GPRBOA-RV method, the step size can be adjusted depending on the process requirements. The part location uncertainties are generated randomly using the position bound ±1 mm and orientation bound ±0.6°.

1) *DOE Experimental Results:* For DOE 3 L, each parameter has three levels. In order to evaluate all possible parameter combinations, $3^3 = 27$ experiments must be performed. Because of the uncertainties and noise in the real applications, one set of 27 experiments is not idea to explore the optimal set of assembly parameters. Hence, 10 sets of 27 experiments (totally 270 experiments) were performed. After processing the collected data, 27 experimental results are shown in Table 3.

After evaluating the results in Table 3, we find that the best set of assembly parameters is [350,20,50] with cycle time 2.3 s and variance 0.09 s. The other two sets of parameters [350,20,75] and [300,20,100] also achieve similar results. These two sets of parameters have the same SF; hence, SF is the most sensitive parameter for this application.

2) *Results of Proposed Method:* The proposed method was implemented without stopping the assembly process. Only are four sets of GPRBOA-RV experimental results shown in this paper (3P means three parameters): 3 L #1, 3 L #2, ML #1 and ML #2.

In the modeling process, we combined two covariance functions in Table 1, i.e., Squared Exponential Function and Gaussian Radial Basis Function. Therefore, three hyperparameters in the GPR model were optimized when constructing the GPR model. From the experimental results shown in Fig. 6, we can see that the assembly process converges after eight experiments for 3 L with a set of optimal assembly parameters [350,20,50]. For ML, optimal sets of parameters are obtained after 10 experiments. However, the optimal set of parameters for the two ML experiments are different, but quite close. This is caused by the system uncertainties and noise. These results also verify that the system uncertainties and noise will affect the optimal parameters. Hence, the optimal parameters should be tuned once the part variations and/or system uncertainties are changed significantly.

The experiments were performed many times with different uncertainties and noise and the results are quite consistent, which means that the proposed method is quite robust to optimize the assembly process parameters.

3) *Discussion:* The number of levels in the DOE method greatly affects the efficiency of the assembly process optimization. For example, if there are M parameters and each parameter has N_i (i=1,...,M) levels, the number of experiments to be performed is $\prod_{i=1}^{M} N_i$ using the full

Table 2
Assembly parameters.

EXPERIMENTS	IF (N)	SS (mm/s)	SF (N)
DOE 3L	50:25:100	250:50:350	5:15:35
GPRBOA-RV 3 L	50:25:100	250:50:350	5:15:35
GPRBOA-RV ML	50:5:100	250:10:350	5:0.5:35

Table 3
DOE 3 L results.

SS (mm/s)	SF (N)	IF (N)	ϕ (s)	ψ (s)
250	35	50	6	0.68
300	35	50	5.5	1.05
350	35	50	5.1	1.67
250	20	50	2.4	0.17
300	20	50	2.3	0.13
350	20	50	2.3	0.09
250	5	50	6.1	0.62
300	5	50	6.1	0.56
350	5	50	6	0.42
250	35	75	5.5	1.18
300	35	75	5.4	0.79
350	35	75	5.9	1.11
250	20	75	2.4	0.2
300	20	75	2.4	0.11
350	20	75	2.4	0.09
250	5	75	6.2	0.52
300	5	75	6	0.56
350	5	75	5.8	0.67
250	35	100	5.6	0.88
300	35	100	6.1	0.45
350	35	100	6.1	0.86
250	20	100	2.5	0.14
300	20	100	2.3	0.1
350	20	100	2.3	0.14
250	5	100	5.9	0.45
300	5	100	5.9	0.58
350	5	100	6	0.49

Note: ϕ and ψ are the mean and variance of the assembly cycle time.

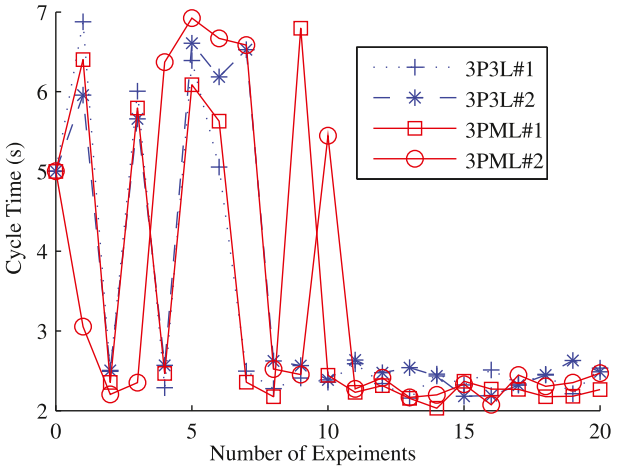


Fig. 6. Four experimental results.

factorial method. This means the number of experiments will increase significantly if the numbers of assembly parameters and parameter values increase.

The proposed method evaluates all parameter values at each iteration; but the number of experiments does not increase significantly even though the levels of the assembly parameters and/or the number of parameters increase. The number of parameter combinations for ML is $11 \times 61 \times 11 = 7381$; however, only are 10 experiments needed to generate a set of optimal parameters.

The identified optimal parameters, cycle time including mean and variance, and the total experimental time are listed in Table 4. Because the implementation of the DOE method was done offline, the data processing time is not included.

From Table 4, we can see that the cycle time and optimal parameters for different methods are quite similar; however, the experimental time to generate a set of optimal parameters is greatly reduced using the proposed method.

Table 4 also shows that the mean cycle time of GPRBOA-RV ML is shorter than that of the DOE method. This is because GPRBOA-RV ML evaluates more parameters, thus, generates more precise results.

3.2. Experiments for torque converter assembly

A torque converter assembly process used to validate the proposed method is shown in Fig. 7. The system consists of an ABB IRB4400 robot with an IRC5 robot controller and an ATI force/torque sensor mounted on the robot end-effector to measure the assembly force/torque. The proposed optimization algorithm was implemented on a computer. After the robot controller collects the data, they are sent to the computer via ethernet. The computer generates a set of new parameters and sends it to the robot controller to perform experiment again.

Torque converters are typically quite heavy, up to 75 kg. There are several individual gear assembly stages in the torque converter assembly process. Because the assembly process is quite difficult and workers have to lift up heavy parts continuously, the manual assembly process could cause injury and other safety issues. The torque converter assembly process includes several steps. Firstly, the shaft in the transmission body should be aligned with the hole in the torque converter. Secondly, the converter must be rotated in order to match gears. Thirdly the converter must be hopped up and down to avoid gear surfaces being stuck together. Lastly, some transmissions have an oil pump. The oil pump typically has two teeth. In order to engage the oil pump, the converter must be rotated more than 180°. which could increase the assembly cycle time.

Table 4
Result comparison among different methods.

Methods	Optimal parameters	Φ (s)	Ψ (s)	Number of experiments	Total time (s)
DOE	[350,20,50]	2.32	0.09	270	1254
L #1	[350,20,50]	2.39	0.13	8	36.61
L #2	[350,20,50]	2.44	0.14	13	53.7
ML #1	[350,24,50]	2.31	0.14	13	54.26
ML #2	[350,23.5,50]	2.23	0.1	13	52.05

Because the transmission body and torque converter have location variation and dimensional uncertainties, assembly failure could happen. The FTT rate of the torque converter assembly process could be decreased from 98% to about 60% or less if the assembly process parameters are not tuned properly.

To avoid the contact friction force and overcome sliding problem between the mating gears, hopping and/or oscillation are typically utilized along the insertion direction. Hence, the complex torque converter assembly process involves several parameters, such as rotation speed, rotation amplitude, search force, hopping force period, insertion distance, insertion force and engaging distance. This means at least seven parameters should be optimized simultaneously. In the experimental system, this torque converter assembly process has three stages; thus, 3 sets of parameters (total 21 parameters) must be optimized.

We simulated the uncertainties and noise of the real assembly processes by adding random torque converter position and orientation variations which were generated randomly based on the maximum position error ±0.6 mm and maximum orientation error ±2.5°. Using the traditional assembly method without online optimization, the assembly process fails to handle the variations. However, using the proposed GPRBOA-RV method, the complex torque converter assembly process was successfully implemented, which means the proposed method can adapt to the uncertainties and variations. Fig. 8 shows one of the examples of random variations added to the torque converter assembly process. In the modeling process, we also combined two covariance functions in Table 1, i.e., Squared Exponential Function and Gaussian Radial Basis Function.

For the first assembly stage, Fig. 9 shows a typical result. After five experiments, the optimal set of parameters was identified. The exploration and exploitation processes are listed in the figure. The production process started after the assembly parameters were optimized. We also tested the assembly process using the optimal parameters. We can see from Fig. 9 that the cycle time is keeping changing. This is caused by the cyclic gear location variation and system uncertainties.

The experimental results for the second and third stages are quite similar to those of the first stage as shown in Figs. 10-11 respectively. The second assembly stage converges much faster because it is easier to be assembled. The third stage is the oil pump assembly. As mentioned before, the torque converter has to be rotated more than 180° in order to match the gear teeth in the oil pump, which requires longer time. Thus, its cycle time is much longer than those of others.

The torque converter assembly process has been tested many times and the results are quite consistent even though we used different torque converters. Hence, the proposed method can optimize the complex torque converter assembly process very efficiently.

4. Conclusion and future work

In this paper, we propose a process parameter optimization method to optimize the performance of complex assembly processes online. A non-parametric model is constructed based on Gaussian Process Regression. The modeling process is performed iteratively using the collected data. At each iteration, a new set of assembly process parameters is generated using Bayesian Optimization Algorithm. To avoid the optimization process converging to a local minimum, a GPRBOA-RV method is developed. By balancing the exploitation and exploration processes, the proposed method can optimize a complex system very

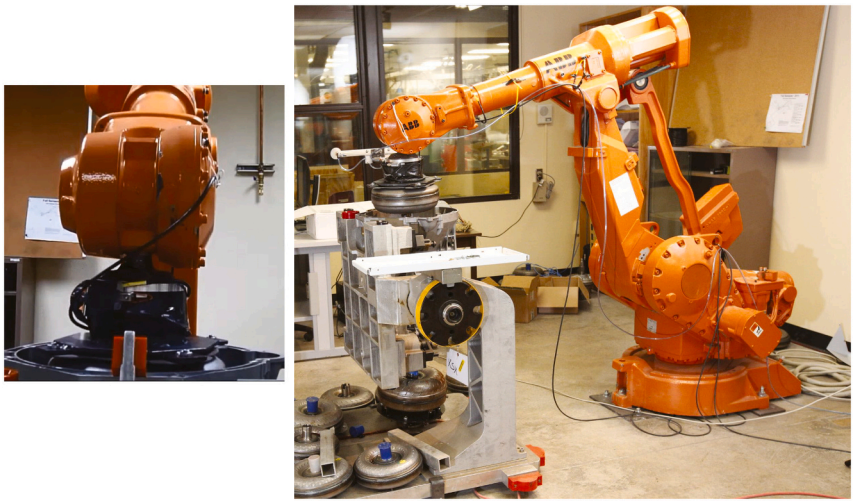


Fig. 7. The torque converter assembly system with force/torque sensor.

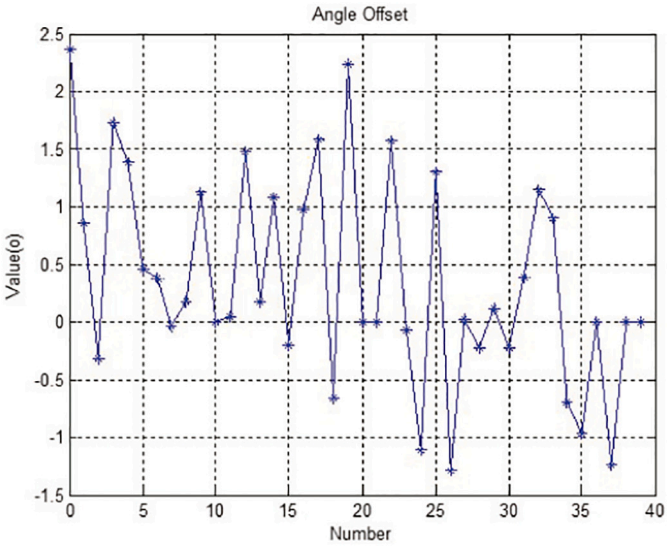


Fig. 8. Random disturbance added to the torque converter assembly process.

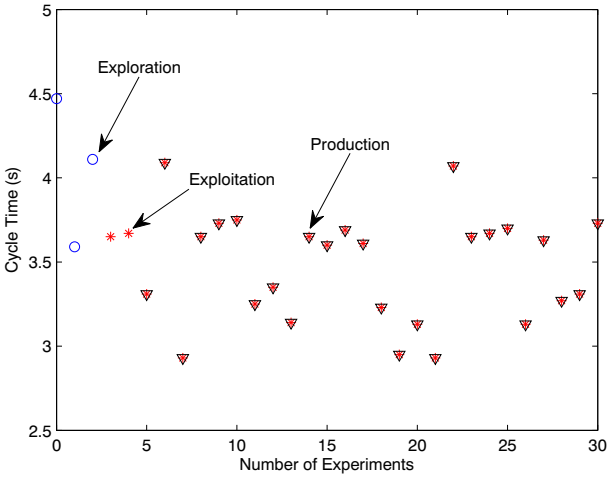


Fig. 9. The first assembly stage: optimal set of parameters was identified using five experiments.

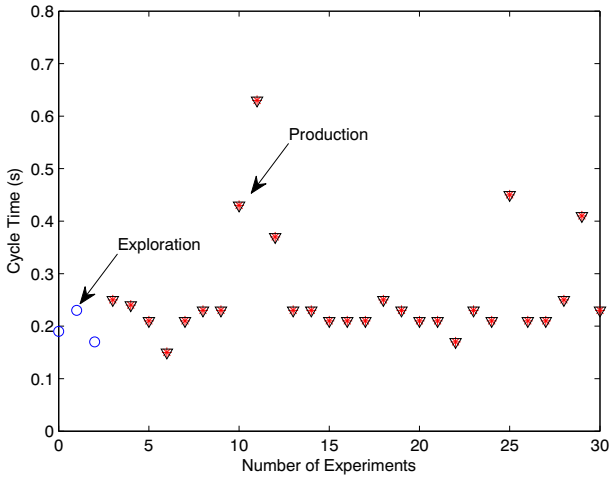


Fig. 10. The second assembly stage: optimal set of parameters was identified using three experiments.

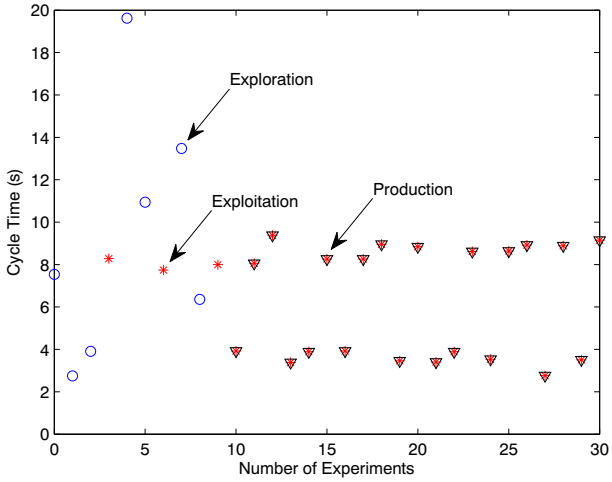


Fig. 11. The third assembly stage (oil pump): optimal set of parameters was identified using ten experiments.

efficiently. Two assembly processes, valve body assembly and torque converter assembly, were implemented to verify the proposed method. Compared with the traditional DOE method, the proposed method can greatly increase the optimization efficiency.

The proposed method can also handle system uncertainties and part dimensional variations, which greatly increases the manufacturing robustness. Moreover the proposed method can optimize the assembly processes using a fewer number of experiments, thus, it can significantly decrease the manufacturing cost and improve the manufacturing efficiency. Hence, the proposed method will advance the state-of-the-art complex manufacturing process optimization technology.

Even though the proposed method can optimize the assembly performance online, it has difficulty to deal with assemblies using parts different batches. We investigated the problem and discussed with the engineers at the Ford Motor Company and found that the variations of the part geometry and dimension are quite large from batch to batch. Therefore, we should re-optimize the assembly process parameters once a new batch is launched. Hence, how and when to start the re-optimization process online needs to be investigated during the production.

Because the GPR modeling method and BOA are very general in system modeling and optimization, we will implement the proposed method in other manufacturing processes such as welding process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This paper is partially supported by ABB Inc.

References

[1] Dietrich F, Buchholz D, Wobbe F, Sowinski F, Raatz A, Schumacher W, et al. On contact models for assembly tasks: experimental investigation beyond the peg-in-hole problem on the example of force-torque maps. Ger Res 2010:2313–8.
[2] Jing W, Newman WS. Improving robotic assembly performance through autonomous exploration. In: Robotics and Automation. 3. 2002 Proceedings ICRA '02 IEEE International Conference on; 2002. p. 3303–8.
[3] Daneshmand M, Noroozi F, Corneanu CA, Mafakheri F, Fiorini P. Industry 4.0 and prospects of circular economy: a survey of robotic assembly and disassembly. CoRR 2021;abs/2106.07270 [Online]. Available: https://arxiv.org/abs/2106.07270.
[4] Li X, Xiao J, Zhao W, Liu H, Wang G. Multiple peg-in-hole compliant assembly based on a learning-accelerated deep deterministic policy gradient strategy. Industrial Robot: the International Journal of Robotics Research and Application 2021;2021/10/12.
[5] Kluz R, Trzepiecinski T. Optimization of selected parameters of modular assembly robot. In: *Theory and practice of industrial and production engineering*, ser. Applied mechanics and materials. vol. 791. Trans Tech Publications Ltd.; 2015. p. 166–73. 10.
[6] Yamanobe N, Fujii H, Maeda Y, Arai T, Watanabe A, Kato T, et al. Optimization of damping control parameters for cycle time reduction in clutch assembly. In: Intelligent Robots and Systems. 2005 (IROS 2005) 2005 IEEE/RSJ International Conference on; 2005. p. 3251–6.
[7] Wei J, Newman W. Improving robotic assembly performance through autonomous exploration. In: Robotics and Automation. 3. Proceedings. ICRA '02. IEEE International Conference on; 2002. p. 3303–8. 2002.
[8] Marvel J, Newman W, Gravel D, Zhang G, Wang J, Fuhlbrigge T. Automated learning for parameter optimization of robotic assembly tasks utilizing genetic algorithms. IEEE International Conference on Robotics and Biomimetics(ROBIO) Feb. 2009:179–84.
[9] Levitin G, Rubinovitz J, Shnits B. A genetic algorithm for robotic assembly line balancing. Eur J Oper Res 2006;168(3):811–25 [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0377221704004874.
[10] Marvel J, Newman W, Gravel D, Zhang G, Fuhlbrigge T. Automated learning for parameter optimization of robotic assembly tasks utilizing genetic algorithms. 2008 IEEE International Conference on Robotics and Biomimetics Feb. 2009:179–84.
[11] Marvel J, Newman W. Accelerating robotic assembly parameter optimization through the generation of internal models. IEEE International Conference on Technologies for Practical Robot Applications(TePRA) Nov. 2009:42–7.
[12] Marvel Jeremy A, Newman Wyatt S. Model-assisted stochastic learning for robotic applications. Automation Science and Engineering, IEEE Transactions on 2011;8(4):835–45.
[13] Gruendling T, Guilhaus M, Barner-Kowollik C. Design of experiment (doe) as a tool for the optimization of source conditions in sec-esi-ms of functional synthetic polymers synthesized via atrp. Macromol Rapid Commun 2009;30(8):589–97. https://doi.org/10.1002/marc.200800738. Online]. Available:.
[14] Antony J. Design of Experiments for engineers and scientists. 1st ed. Elsevier; 2003.
[15] Gravel D, Maslar F, Zhang G, Nidamarthi S, Chen H, Fuhlbrigge T. Toward robotizing powertrain assembly. In: Intelligent Control and Automation. WCICA 2008. 7th World Congress on, June 2008; 2008. p. 541–6.
[16] Zhang G, Bell A, Martinez C. On-pendant robotic assembly parameter optimization. In: 7th World Congress on Intelligent Control and Automation. 2008; 2008. p. 547–52 [Online]. Available: http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=4592982.
[17] Gravel D, Zhang G, Bell A, Zhang B. Objective metric study for DOE-based parameter optimization in robotic torque converter assembly. The IEEE/RSJ International Conference on Intelligent Robots and Systems Oct. 2009:3832–7.
[18] Zhang B. Robotic force control assembly parameter optimization for adaptive production. In: IEEE International Conference on Robotics and Automation, Shanghai, China; 2011.
[19] Gravel D, Zhang G, Bell A, Zhang B. Robot learning and self optimization of process parameters. In: *ASME/ISCIE International Symposium on Flexible Automation*, St. Louis, MO; 2012. p. 1–7.
[20] Tomasic I, Andersson A, Funk P. Mixed-effect models for the analysis and optimization of sheet-metal assembly processes. IEEE Transactions on Industrial Informatics 2017;13(5):2194–202.
[21] Williams CRC. Gaussian processes for regression. Adv Neural Inf Proces Syst 1996;8:598–604.
[22] Rasmussen CE. Gaussian processes for machine learning. In: Adaptive computation and machine learning series. MIT Press; 2006.
[23] Grancharova A, Kocijan J, Johansen TA. Explicit stochastic predictive control of combustion plants based on gaussian process models. Automatica. In: Stochastic Modelling, Control, and Robust Optimization at the Crossroads of Engineering, Environmental Economics, and Finance¡/ce:title¿ 44; 2008. p. 1621–31. no. 6. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0005109808001477.
[24] Kocijan J, Murray-Smith R. Nonlinear predictive control with a gaussian process model. In: Murray-Smith R, Shorten R, editors. *Switching and learning in feedback systems*, ser. Lecture notes in computer science. 3355. Berlin Heidelberg: Springer; 2005. p. 185–200.
[25] Cheng L, Ramchandran S, Vatanen T, Lietzén N, Lahesmaa R, Vehtari A, et al. An additive gaussian process regression model for interpretable non-parametric analysis of longitudinal data. Nat Commun 2019;10(1):1798.
[26] Wan J, McLoone S. Gaussian process regression for virtual metrology-enabled run-to-run control in semiconductor manufacturing. IEEE Trans Semicond Manuf 2018;31(1):12–21.
[27] Jadaliha M, Xu Y, Choi J, Johnson N, Li W. Gaussian process regression for sensor networks under localization uncertainty. Signal Processing, IEEE Transactions on 2013;61(2):223–37.
[28] Richardson RR, Birkl CR, Osborne MA, Howey D. Gaussian process regression for in-situ capacity estimation of lithium-ion batteries. IEEE Transactions on Industrial Informatics 2018;no. 99:1. vol. PP.
[29] Schneider M, Ertel W. Robot learning by demonstration with local gaussian process regression. In: Intelligent Robots and Systems (IROS). 2010 IEEE/RSJ International Conference on; 2010. p. 255–60.
[30] Vasudevan S, Ramos F, Nettleton E, Durrant-Whyte H. Gaussian process modeling of large-scale terrain. Journal of Field Robotics 2009;26(10):812–40.
[31] Frank B, Stachniss C, Abdo N, Burgard W. Using Gaussian process regression for efficient motion planning in environments with deformable objects. In: Automated action planning for autonomous Mobile Robots '11; 2011.
[32] Gregorcic G, Lightbody G. Gaussian process approach for modelling of nonlinear systems. Eng Appl Artif Intell 2009;22(4–5):522–33 [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0952197609000268.
[33] Yang K, Gan S Keat, Sukkarieh S. A gaussian process-based rrt planner for the exploration of an unknown and cluttered environment with a uav. Adv Robot 2013;27(6):431–43. https://doi.org/10.1080/01691864.2013.756386 [Online]. Available:.
[34] Khan N, Khan N, Goldberg DE, Goldberg DE, Pelikan M, Pelikan M. Multi-objective bayesian optimization algorithm. Proceedings of the Genetic and Evolutionary Computation Conference 2002. Tech. Rep.
[35] Pelikan M. Bayesian optimization algorithm. In: *Hierarchical Bayesian optimization algorithm*, ser. Studies in fuzziness and soft computing. vol. 170. Berlin Heidelberg: Springer; 2005. p. 31–48.
[36] Hasegawa Y, Iba H. A bayesian network approach to program generation. Evolutionary Computation, IEEE Transactions on 2008;12(6):750–64.
[37] Lima C, Lobo F, Pelikan M, Goldberg D. Model accuracy in the bayesian optimization algorithm. Soft Comput 2011;15(7):1351–71. https://doi.org/10.1007/s00500-010-0675-y [Online]. Available:.
[38] Zhang Y, Apley DW, Chen W. Bayesian optimization for materials design with mixed quantitative and qualitative variables. Sci Rep 2020;10(1):4924.
[39] Wu J, Chen X-Y, Zhang H, Xiong L-D, Lei H, Deng S-H. Hyperparameter optimization for machine learning models based on bayesian optimization. Journal of Electronic Science and Technology 2019;17(1):26–40.
[40] Yang J, Xu H, Pan L, Jia P, Long F, Jie M. Task scheduling using bayesian optimization algorithm for heterogeneous computing environments. Appl Soft

Comput 2011;11(4):3297–310 [Online]. Available: http://www.sciencedirect.com/science/article/pii/S1568494610003078.
[41] Garnett R, Osborne MA, Roberts SJ. Bayesian optimization for sensor set selection. In: *Proceedings of the 9th ACM/IEEE international conference on information processing in sensor networks*, ser. IPSN '10. New York, NY, USA: ACM; 2010. p. 209–19.
[42] Yang J, Xu H, Jia P. Task scheduling for heterogeneous computing based on bayesian optimization algorithm. In: Computational Intelligence and Security. 1. 2009 CIS '09 International Conference on; 2009. p. 112–7.
[43] Forrester AI, Keane AJ. Recent advances in surrogate-based optimization. Prog Aerosp Sci 2009;45(1–3):50–79 [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0376042108000766.
[44] Jin Y. Surrogate-assisted evolutionary computation: recent advances and future challenges. Swarm and Evolutionary Computation 2011;1(2):61–70 [Online]. Available: http://www.sciencedirect.com/science/article/pii/S2210650211000198.
[45] Brochu E, Cora VM, de Freitas N. A tutorial on bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. CoRR 2010;abs/1012.2599.
[46] Jalali A, Azimi J, Fern X. Exploration vs exploitation in bayesian optimization. CoRR 2012;abs/1204.0047.
[47] Bardenet R, KŠSgl B. Surrogating the surrogate: Accelerating gaussian-process-based global optimization with a mixture cross-entropy algorithm. In: ICML '10; 2010. p. 55–62.
[48] Chen H, Bowels S, Zhang B, Fuhlbrigge T. Controller parameter optimization for complex industrial system with uncertainties. Meas Control 2019;52(7–8):888–95. https://doi.org/10.1177/0020294019830108.
[49] Fern lan, Tadepalli P. A behavior based kernel for policy search via Bayesian optimization. In: Proceedings of the ICML 2011 workshop: Planning and acting with uncertain model, Bellevue, WA; 2011.
[50] Marchant R, Ramos F. Bayesian optimisation for intelligent environmental monitoring. In: Intelligent Robots and Systems (IROS). 2012 IEEE/RSJ International Conference on; 2012. p. 2242–9.
[51] Marvel J, Falco J. Best practices and performance metrics using force control for robotic assembly [Online] Available: http://nvlpubs.nist.gov/nistpubs/ir/2012/NIST.IR.7901.pdf; 2012.
[52] Garivier A, Moulines E. On upper-confidence bound policies for switching bandit problems. In: *Proceedings of the 22nd International Conference on Algorithmic Learning Theory*, ser. ALT'11. Berlin, Heidelberg: Springer-Verlag; 2011. p. 174–88.
[53] Hawe G, Sykulski J. An enhanced probability of improvement utility function for locating pareto optimal solutions. In: 16th conference on the computation of electromagnetic fields COMPUMAG. vol. 3. RWTH Aachen University; 2007. p. 965–6. event Dates: 25? 28 June 2007. [Online]. Available: http://eprints.soton.ac.uk/264368/.
[54] Kleijnen J, Beers W, Nieuwenhuyse I. Expected improvement in efficient global optimization through bootstrapped kriging. J Glob Optim 2012;54(1):59–73. https://doi.org/10.1007/s10898-011-9741-y [Online]. Available:.
[55] Raibert MH, Craig JJ. Hybrid position/force control of manipulators. J Dyn Syst Meas Control 1981;103(2):126–33. 06.